

Test Documentation: Battery and Power Management

Module: Battery & Power Management

Functionality: Power Saving Mode

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_BPM_001	Verify enabling Power Saving Mode manually.	Device battery is above 20%.	1. Go to Settings → Battery. 2. Toggle Power Saving Mode to ON.	1. The battery icon in the status bar must turn yellow. 2. Screen brightness should decrease slightly. 3. Auto-screen off time should drop to 15 seconds.	Pass
TC_BPM_002	Verify auto-turn off functionality of Power Saving Mode.	Power Saving Mode is ON. “Turn off when charged” is enabled.	1. Connect the device to a charger. 2. Allow the battery to charge above the threshold (e.g., 60% or 90% depending on OS version).	Power Saving Mode must automatically toggle OFF, and the battery icon should return to its default color.	Pass
TC_BPM_003	Verify auto-turn on at a specified battery level.	Power Saving Mode is OFF. “Turn on at specified battery level” is set to 20%.	1. Drain the device battery naturally or via a heavy benchmark app to reach 20%.	The system must prompt a warning and automatically enable Power Saving Mode exactly at the 20% mark.	Pass

Functionality: Super Power Saving Mode

Note: This mode heavily restricts background activities, limits CPU performance, and allows only a maximum of 6 apps on a dark home screen.

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_BPM_004	Verify enabling Super Power Saving Mode.	Device is in normal operating mode.	1. Go to Settings → Battery → Advanced/More Settings. 2. Toggle Super Power Saving Mode to ON. 3. Tap “Turn On” on the confirmation prompt.	The UI must transition to a simplified dark mode screen. Only Phone, Messages, Contacts, and up to 3 custom apps should be visible. System animations should be disabled.	Pass
TC_BPM_005	Verify adding and removing apps in Super Power Saving Mode.	Super Power Saving Mode is active.	1. Tap the Add (+) icon on the simplified home screen. 2. Select an app (e.g., WhatsApp). 3. Long press the WhatsApp icon and select Remove.	1. The selected app must be added to the allowed list and launch successfully. 2. The removed app must disappear from the screen, making the slot empty again.	Pass
TC_BPM_006	Verify system behavior for background notifications in Super Power Saving Mode.	Super Power Saving Mode is active. Wi-Fi/Mobile Data is ON.	1. Send a test email or a third-party app notification (e.g., Telegram) from another device.	The notification must NOT be received in real-time unless the specific app is added to the allowed 6-app list, confirming background data restriction.	Pass
TC_BPM_007	Verify exiting Super Power Saving Mode.	Super Power Saving Mode is active.	1. Tap the Exit icon (door icon) at the top left/right corner. 2. Confirm the exit prompt.	The device must smoothly transition back to the standard ColorOS Home Screen, restoring all UI elements, wallpapers, and background processes.	Pass

Functionality: Advanced Battery Settings & Optimization

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_BPM_008	Verify High Performance Mode functionality.	Device is normal. Battery is above 50%.	1. Go to Settings → Battery → Advanced Settings. 2. Toggle High Performance Mode to ON.	A permanent notification/status bar icon must appear indicating the mode is active. CPU/GPU benchmarks should show higher scores, and power consumption should increase.	Pass
TC_BPM_009	Verify Optimized Night Charging (AI Battery Protection).	“Optimized Night Charging” is toggled ON.	1. Plug the device into the charger late at night (e.g., 11:30 PM). 2. Monitor the charging graph via settings or adb logs over 8 hours.	The battery must charge to 80% and pause. It should intelligently resume and hit 100% just before the user’s usual wake-up time (e.g., 6:00 AM).	Pass
TC_BPM_010	Verify App Battery Management (Background restrictions).	A third-party heavy app (e.g., Facebook) is installed.	1. Go to Settings → Battery → App battery management. 2. Select Facebook. 3. Select Don’t run in background. 4. Open Facebook, then go to the Home screen and wait 5 minutes.	The OS must forcefully kill the Facebook background process. No notifications from Facebook should arrive until the app is manually opened again.	Pass